NIPUN KAKKAR

AI Engineer and Data Scientist

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PROFESSIONAL SUMMARY

- AI Engineer & Data Scientist with 8+ years of experience across Optum, IBM, Deloitte, Adcuratio and Infosys.
- Versatile and highly skilled AI developer with practical exposure, certifications and proven ability to lead teams, own complete product delivery, and drive business value through technologies like ML, Deep learning, Recommender Systems and Agentic AI.
- Experienced in domains like Banking, Finance, Insurance (BFSI), Healthcare and Media & Advertisement.

WORK EXPERIENCE

1. **Optum** Global Solutions India Pvt. Ltd. (June 2026 - Present Full Time)

    - Working on a fast paced project in Insurance Grievance to automated case closing letter which uses Multi-Agent Reflect Rag application, which includes healthcare domain knowledge, understanding manual process, deep data and database understanding, Traditional Classification Model, LLM as a judge and other state of the art technology and logical reasoning.

2. **IBM** India Pvt. Ltd. (Oct 2023 – May 2026 Full Time)

    - Led and owned development of enterprise AI solutions including LLM-powered Chatbot, intelligent Ticket Solution Platform, COBOL legacy-code Functionality Extraction using GenAI, RAG Pipelines, Multi-Agent Orchestration.
    - Worked across full AI lifecycle including architecture design, model orchestration, vector retrieval pipelines, deployment, backend Api development and production integration.

3. **Deloitte USI** (May 2022 – Oct 2023 Full Time)

    - Got appreciation from Director of Service Line for designing and delivering an end-to-end Python automation platform that reduced nearly 50% of manual QA operational effort.
    - Early Adopter of GenAI and LLMs for Data Governance initiatives, ETL and other projects upon launch of ChatGPT.

4. **Adcuratio** Media India Pvt. Ltd. (Nov 2020 – April 2022 Full Time)

    - Led the most important project of rapidly growing company by building a hybrid and scalable personalized OTT Recommendation Engine serving US based streaming app using Neo4j, Deep Learning, Content Based Filtering and Collaborative Filtering.
    - Managed end-to-end deployment including backend services, server, GPU environments, database infrastructure, and integration support for frontend applications.
    - Acted as lead Data Scientist, involved in multiple major and minor projects and 30+ candidate interviews.

5. **Infosys** Ltd. (Jan 2018 – Nov 2020 Full Time)

    - Started with training in technologies like Python, SQL and Data Structures & Algorithms then transitioned from QA into AI/ML focused roles quite early in the career.
    - Worked on machine learning and data science projects involving exploratory data analysis, visualization, feature engineering, customer segmentation and predictive modeling.

CERTIFICATIONS

- Google Cloud Certified Professional Machine Learning Engineer
- Microsoft Azure Certified AI Engineer Associate
- IBM Certified Data Science Professional
- Infosys Certified Machine Learning Professional
- Infosys Certified Python Programmer
- Infosys Certified Database and SQL Professional

- IgnisWay Certified Web Developer

EDUCATIONAL QUALIFICATIONS

Course/Grade	Institution	Completion Year	Performance
• Bachelor of Computer Applications	St. Xavier's College, Jaipur (RU)	2017	69.23%
• 12th Grade	Nosegay Public School (CBSE)	2014	90.40 %

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE SKILLS

- **Generative AI and Agentic AI -** Transformers, Vision Transformers, LangChain, LLM, Prompt Engineering, RAG Pipeline, Agentic RAG, Vectorless RAG, Vector Databases, Chatbots, AI Agents, CrewAI, LangGraph, Multi-Agent Architectures, Agentic Reflect Loop, Loop Engineering, LLM Finetuning, Stateful AI Workflows, Middlewares, Guardrails, Load Balancing, LLM Routing, Gateways, Human-in-the-Loop Systems, MCP

- **Recommender Systems -** Content Based Filtering, Collaborative Filtering, Hybrid Approach, Neo4j, A/B Testing

- **Natural Language Processing** - Bag of Words, TFIDF, Bidirectional LSTM, Cosine Similarity, Word2Vec, Bert

- **Machine Learning and Deep Learning** – Exploratory Data Analysis, Regression, Classification, Clustering, DBScan Clustering, Random Forests, Extra Trees, Boosting Algorithms, Artificial Neural Networks, CNN, RNN, GNN, Computer Vision, Object Detection, YOLO

- **Cloud Infrastructure and CI/CD Pipeline** - MLOps, LLMOps, Azure, GCP, AWS, Heroku, Railway, GitHub, Dockers, GitHub Action, Airflow, DagsHub, MLFlow, Jenkins, CircleCI, DVC, FastAPI, MySQL, Oracle, MongoDB, PostgreSQL, Neo4j, MS SQL Server

MAJOR PROJECTS

1. **Grievance Closing Letter Automation Project (Multi-Agent Reflect Architecture and Corrective Rag Pipeline)**

    - A fast paced and very important project in healthcare domain, grievance involves very unstructured and large amount of data when user raises a case or complaint.
    - The complaint is written in unstructured form in initial correspondence
    - The project involves understanding the raw text of init corr with OCR
    - Then identifying multiple issues and their categories namely RFC, DRFC, Issue Description and evidence
    - After understanding it we have to decide which issue is important enough to be cited in the closing letter and for those issue corresponding Chapter and Sections needs be found from Evidence of Coverage Document.
    - Not only Chapter and Section but based on Issue of the case the releavant text of that issue from EoC should also be printed in the Closing Letter.
    - For both the things multiple different technologies like Multi-Agentic Reflect Loop, XGBoost to improve decision accuracy, SME feedback with sample data, historical database data, LLM-as-a-judge and many others.
    - At the end using LLM, Corrective Vectorless RAG implementation, SME feedback, LLM-as-a-judge the project was shipped within the deadlines.
    - At the end Decision Accuracy and Section Cited Accuracy both combined came out to 95%

2. **Enterprise AI Ticket Solution Platform (Using Multi-Agent Architecture)**

    - Designed and implemented an LLM-powered ticket classification system for SAP support where users submitted textual issue descriptions and error screenshots via UI.
    - Extracted structured data from screenshots (e.g., User ID, environment, error message) using OCR, and validated it against the user-written issue using LLM prompts.
    - Fetched additional user information, from UI payload (e.g., user email, logged in userid), from SAP APIs (e.g., validity expire, lock status code).

- Using LLMs checked for quality and size of image, issue description input for consistency and correctness and then generating final summary of data.
- Used multi-agent logic with LLM to classify the issue into predefined types (e.g., Password Incorrect, Invalid Login).
- Developed separate UI views: one for users to raise and track tickets, and another for backend agents to modify and close them. For all this, fetching of data, Update/Patch etc. endpoints using FastAPI were created
- Based on the classified issue (e.g., Password forgotten/incorrect), corresponding actions like password reset were automatically triggered via APIs, and ticket status was updated to closed_complete in the ServiceNow backend. Thus, reducing manual intervention and accelerating resolution time.

3. GenAI Support Assistant Chatbot for IBM IQP Product (Using GenAI and RAG Pipeline)

- Designed and led a full-scale GenAI-powered multi-tier chatbot for an IBM product with a team of 4 members.
- The first layer used intent-entity based RASA NLU trained with YAML configs and curated Q&A data. It was tested with alternate questions, synonyms of keywords etc. If RASA failed to respond adequately, fallback mechanisms triggered NLP-based cosine similarity logic.
- Final fallback leveraged a RAG pipeline powered by LLMs like Llama and Granite through the IBM Watsonx platform using ChromaDB for vector storage and retrieval of product documentation in PDF format. Users could raise JIRA tickets if unsatisfied with responses.
- Achieved all the objectives of creating Chatbot including helping users with queries, navigating resources of the product along with fast and reliable response and all this while minimizing LLM-related costs.

4. Movie and Show Recommendation System (Using Neo4j, Content Based & Hybrid Filtering, NLP and Bert)

- Built a unified recommendation engine for an OTT aggregator app that allowed users in the U.S. to watch content from multiple streaming services (e.g., Netflix, Prime) on a single platform.
- In this app user could watch shows and movies available everywhere on one single place.
- Collected and integrated large-scale movie/show metadata using TMDB, IVA APIs and other sources by web scrapping and structured data ingestion.
- Stored data in MongoDB for structured access, PostgreSQL for tabular queries, and Neo4j for modeling relationships via knowledge graphs (e.g., genres, actors, directors).
- Designed content-based filtering using knowledge graphs and cosine similarity based on user likes/dislikes and watch history.
- Enhanced recommendations using collaborative filtering (user-user and item-item similarity) and social circle behavior (e.g., friend viewing patterns).
- Deployed end-to-end solution on self-managed Linux servers, including full backend setup, GPU and Databases setup, model endpoints, and integration support for frontend developers.
- Dramatically improved user engagement and platform retention using personalized discovery across platforms.

5. Advertisement Watch Prediction Major Project (Using Machine Learning and Deep Learning)

- Led a classification project to predict whether users would watch a particular ad or not, using historical viewing data and program metadata (e.g., channel name, genre, time slot, weekend flag). Built the entire ML pipeline from complex data joins and cleaning, to EDA and feature engineering using Pandas, NumPy and Seaborn & Matplotlib.
- Created custom scripts for data logging, preprocessing, and batch testing.
- Benchmarked multiple models including Logistic Regression, Random Forest, RNN, LSTM, and finally achieved ~80% accuracy using CatBoost.